English_Amharic Medical Image Caption and Speech Generation for Visually Impaired Patients Using Vision Transformer Fused with Generative AI

Dawit Shibabaw[1], Tesfa Tegegne[2], Vukosi Marivate[2], munir.awol[4]

Abstract

Access to medical information is critical for healthcare equity, particularly for visually impaired individuals and low-resource language speakers.
The model is designed to generate detailed captions for radiologists, translate into Amharic for most patients accessible, and speech generation for visually impaired patients.
The model accepts medical images as input, processes them through the models, outputs textual captions, translates them into Amharic, and generates speech.
GitHub repository: Click here

Introduction

- Image captioning is a process of creating human-understandable language descriptive sentences for a given image; these sentences are referred to as the image's caption.
- Radiologists often review medical images, such as X-rays, MRIs, and CT scans, in their practice and describe their findings and impressions in radiology reports.
- The process of interpreting medical images and synthesizing the information they contain **tedious**, **time-consuming**, and **prone to mistakes**, which frequently acts as a bottleneck in the clinical diagnosis procedure.
- In this study, developing model fuss that can generate captions, translate into Amharic and speech from images.

Objectives

- Develop a Medical image caption generate system.
- Translate into Amharic captioning system.
- Convert captions into speech.

Methods

Dataset: This study used secondary data, from the Tikur Anbessa Hospital cancer center, Addis Ababa University, Ethiopia.
We have used a total dataset of 3410 images. In this study x-ray, MRI, and CT scan images are included for patients diagnosed with liver, lung, breast, colorecatal, and cervical cancers. A data collection workflow and sample image are presented in Fig 1.

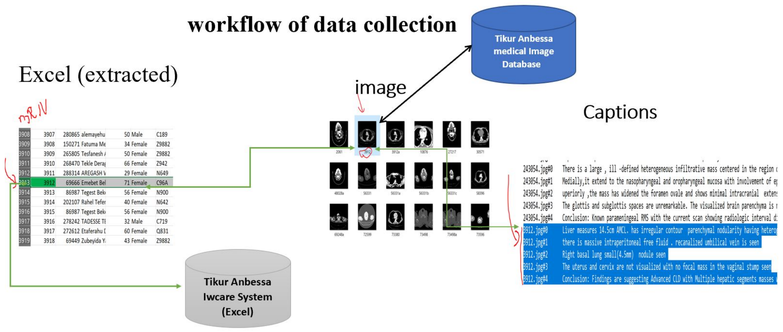


Fig 1: Data collection workflow

Mapping text and image data: a single image contains 2–10 sentences. For example, if Medical record Number(MRN) is 100522 we name it specific image as 100522.jpg, then for a list of sentences/captions for this image `100522.jpg#1`, `100522.jpg#2`, `100522.jpg#3` etc. until cations ends as shown in Fig 2. if not remove captions (or images) that do not have corresponding images (or captions). as shown in Fig 2 take 100522.jpg `image_id` and take a list of caption's `caption_id` such as 0, 1, 2, 3, 4 and caption.

	image_id	caption_id	caption
0	100522.jpg	0	startseq There is right large heterogeneously ...
1	100522.jpg	1	startseq There is right superior sulcus mass l...
2	100522.jpg	2	startseq There is middle lobe bronchiectasis a...
3	100522.jpg	3	startseq Conclusion Spinal canal narrowing as ...
4	100522.jpg	4	startseq Conclusion Known cervical ca with mul...

Fused Models

- VIT-GPT2 (Transformer-fused GenAI)
- VIT-LLaMA2 (Transformer-fused GenAI)
- VGG16-LSTM (Baseline)
- **Audio Generation:** Google TTS + Tralating into Amharic text.

Architecture

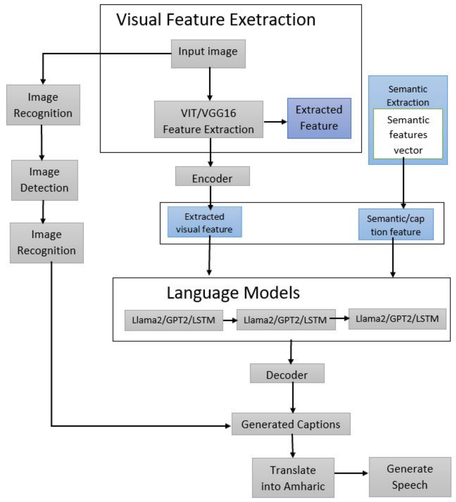


Fig 3: System Architecture

Results and Discussion

we have employed model fusion such as Generative pretrained transformer-2(GPT-2) fussed with Vision transformer(Vit), Llama2 fused with VIT, and VGG16 Fussed with Long short term(LSTM).

Fused models	BLEU	ROUGE-2	ROUGE-L	ROUGE-1	METEOR	CIDEr
VIT-GPT-2	0.543	0.450	0.345	0.254	0.524	0.424
VIT-LLaMA2	0.633	0.250	0.445	0.154	0.324	0.236
VGG16-LSTM	0.342	0.320	0.130	0.163	0.264	0.425

Table: The overall result of the experiment

Discussion:
The model is designed to generate detailed captions for radiologists, translate generated captions into local languages to access their medical result by their mother tongue, and speech for visually impaired patients.

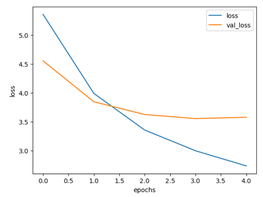


Fig 4: training and validation loss.

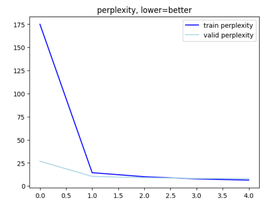


Fig 5: perplexity, lower=better.

EarlyStopping This stops training when the validation loss stops improving, to prevent overfitting and underfitting.
L2 regularization, also known as weight decay or ridge regularization, is a method that penalizes big weights in the loss function in order to reduce model complexity and avoid overfitting.
Perplexity: Measures how well a model predicts a sequence of text. It is formally defined as the exponential of the average negative log-likelihood of a sequence. Mathematically:

$$\text{Perplexity} = 2^{-\frac{1}{N}\sum_{i=1}^{N}\log_2 P(w_i)}$$

Where:
- N: The total number of words (or tokens) in the sequence.
- $P(w_i)$: The probability of the i-th word in the sequence, as predicted by the model.

Caption Quality Comparison with [actual, model and BLEU score]

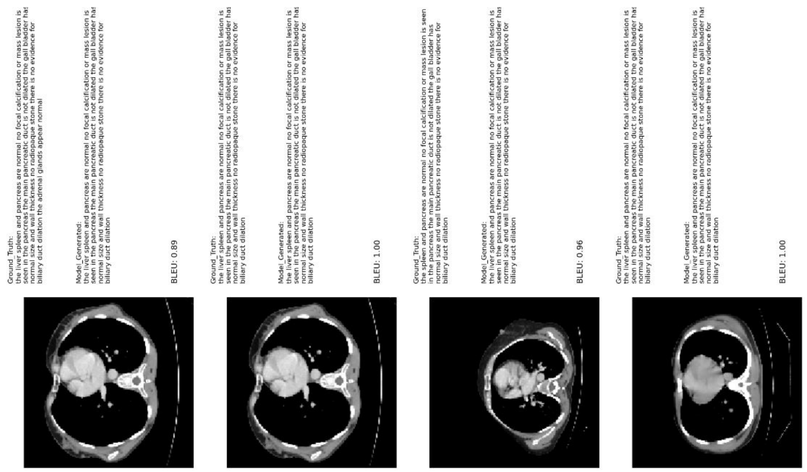
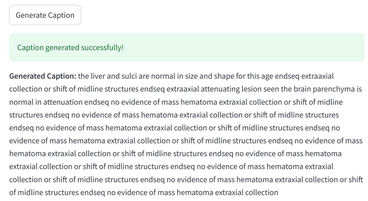


Fig 4: Caption quality comperison

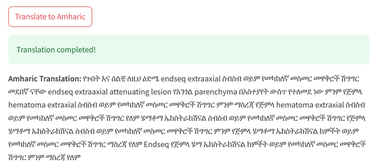
We compared the performance of Generative AI such as ChatGPT-4, Gemini 2.5, and the proposed model fusion (ViT-LLaMA2), GenAI able to generate grammatically accurate captions while score zero BLEU compared to actual captions.

Demonstration:Step1-Step4_Outputs

step1: GUI accept image



step2: Generated Caption (English)



step3: Caption Translated into Amharic

step4: Generated Speech_Visualized Sound Wave

Conclusion

- VIT-LLaMA2 best suited for bilingual captioning.
- Amharic audio greatly enhances accessibility.
- Proposed system shows promise in local health settings.

The system is deployed as a user-friendly application that accepts medical images as input, processes them through the models, outputs textual captions, translates generated caption into Amharic, and speech.